

Akhila Kyatham

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SUMMARY

Data Engineer with 4+ years of experience building and optimizing large-scale ETL pipelines, data warehouses, and real-time streaming systems across financial and IT domains. Skilled in Python, PySpark, SQL, dbt, Spark, Kafka, Snowflake, Databricks, AWS, Azure, and data lake architectures. Proven ability to process multi-terabyte data workloads, improve pipeline performance, reduce costs, and deliver accurate, timely insights. Strong background in ML-based anomaly detection, predictive analytics, and automated KPI dashboards, with solid expertise in Agile, orchestration, and delivering scalable, reliable data engineering solutions.

TECHNICAL SKILLS

Programming & Analytics: Python (Pandas, NumPy, Scikit-learn), PySpark, SQL, Bash, REST APIs, JSON

Machine Learning & Statistical Modeling: Supervised & Unsupervised ML, Feature Engineering, Model Evaluation, A/B Testing, Experimentation Design, Time-Series Analysis, Embeddings Processing, LLM Data Preparation, ML Pipelines

Data Wrangling & Big Data: Data Cleaning & Preprocessing, Spark, PySpark, Databricks, Hive, Hadoop, ETL/ELT Workflows, Parquet/ORC/Avro, Data Lakes (S3, GCS, ADLS)

Cloud & MLOps (Mid-Level DS Relevant): AWS (S3, Athena, Glue), GCP (BigQuery, Dataflow), Azure (Data Factory, Synapse), Model Deployment Basics, Docker (light), CI/CD (GitHub Actions)

Data Warehousing & Analytics Engineering: Dimensional Modeling (Star/Snowflake), Fact/Dimension Tables, Data Marts, dbt, Semantic Layer, Metadata Management, SQL Optimization

Real-Time & Event-Driven Analytics: Kafka, Kinesis, Pub/Sub, Streaming ETL, Micro-batch Processing (optional but valuable for mid-level DS roles)

Data Quality & Governance: Data Validation, Great Expectations, Data Lineage, Data Catalogs, Logging & Monitoring

Visualization & Business Intelligence: Tableau, Power BI, Looker, Quick Sight, KPI Dashboarding, Analytical Storytelling

Soft Skills: Cross-Functional Collaboration, Stakeholder Communication, Problem-Solving, Agile/Scrum, Requirements Gathering, Documentation

WORK EXPERIENCE

UnitedHealth Group - USA | Data Engineer

September 2024 – Present

- Designed and deployed 15+ dbt models and 10+ macros with incremental and modular architecture, boosting ETL throughput for 5+ TB of financial and healthcare data daily and accelerating delivery to 12 downstream warehouses and analytics dashboards.
- Built high-performance Spark and PySpark ETL jobs processing 200+ daily pipelines, reducing execution latency by 25% while optimizing cluster resource utilization, enabling faster generation of financial reports, risk analytics, and healthcare operational insights.
- Implemented serverless streaming pipelines using AWS Lambda & Kinesis to handle 1M+ events/day from financial transactions and healthcare operational systems, cutting operational overhead by 50% and reducing monthly cloud costs by \$12K.
- Developed ML-based anomaly detection models for ETL workflows, proactively identifying inconsistencies across 10+ critical financial and healthcare sources, reducing downstream errors by 80% and ensuring high-quality, compliant data for reporting.
- Optimized Spark executor memory configurations and PySpark transformations, decreasing overall job runtime by 20%, improving large-scale financial and healthcare reporting pipelines, and reducing resource bottlenecks.
- Integrated REST APIs and JSON-based ingestion pipelines from financial and operational healthcare applications enabling near real-time updates and powering dashboards with sub-hour refresh cycles for enterprise stakeholders.
- Automated data quality checks and validation using Great Expectations, ensuring 95% accuracy in key financial and healthcare metrics, reducing manual intervention, and enabling proactive monitoring of pipeline health.
- Collaborated with cross-functional teams and business stakeholders to define data requirements, KPIs, and governance standards, accelerating the delivery of actionable insights to support strategic decision-making across finance and healthcare operations.

Zensar Technologies – India | Data Engineer

August 2020 – July 2023

- Developed and optimized Azure Data Factory ETL pipelines processing over 2 TB of enterprise data per week, improving data flow efficiency by 30%, enabling faster and more reliable IT reporting and analytics across multiple business units.
- Created Airflow DAGs for the orchestration of 50+ pipelines, providing a centralized monitoring framework, reducing job failures by 40%, and improving pipeline reliability through automated retries, logging, and alerting mechanisms.
- Configured Kafka topics and partitions for real-time event ingestion, increasing message throughput by 20% and ensuring reliable, low-latency data delivery for critical IT services and enterprise applications.
- Designed and implemented Snowflake schemas and data marts, optimizing storage usage to reduce costs by 15% while maintaining high-speed query performance for complex multi-source analytics workflows.
- Integrated Spark ML workflows for predictive analytics, including customer churn forecasting and anomaly detection, enhancing delivery of actionable insights by 30% and improving data-driven decision-making for IT and business stakeholders.
- Developed 10+ Power BI dashboards on Synapse data, automating KPI tracking and drill-down analytics to support executive-level strategic planning and operational monitoring across departments.
- Optimized SQL queries and PySpark ETL jobs, improving processing efficiency by 50%, reducing system resource consumption during peak loads, and ensuring the timely availability of high-quality data for downstream applications.
- Implemented comprehensive data governance and metadata tracking, improving data lineage visibility, ensuring regulatory compliance, and enhancing reliability for downstream reporting teams and enterprise analytics consumers.

PROJECTS

Real-Time Streaming Data Platform with ML Anomaly Detection ([GitHub](#))

- Built a real-time streaming pipeline using Python, simulating continuous data ingestion and event-driven processing
- Applied Isolation Forest for anomaly detection, identifying irregular patterns in streaming data in near real-time
- Designed modular pipeline stages (ingestion, processing, validation, analytics), improving system structure and debuggability
- Mapped system architecture to AWS services (Kinesis, S3, Lambda), demonstrating scalability for cloud deployment
- Engineered continuous data flow processing instead of batch workflows, improving understanding of real-time analytics systems

AI Data Validation Platform ([GitHub](#))

- Developed a validation framework for ETL pipelines, ensuring data accuracy across ingestion and transformation stages
- Implemented validation checks (schema, row count, nulls, source-to-target comparison), improving data consistency and reliability
- Built reusable validation modules using Python and Pandas, reducing manual validation effort during testing workflows
- Simulated pipeline failure scenarios and added validation logic, enabling early detection of data issues before downstream impact
- Designed scalable validation workflows aligned with production data systems, improving pipeline monitoring and reliability

End-to-End Data Warehouse & Analytics Platform ([GitHub](#))

- Developed an end-to-end ETL pipeline using Python and SQL to ingest, cleanse, and transform raw sales data into analytics-ready datasets for business reporting.
- Designed dimensional data models with fact and dimension tables, enabling efficient querying and supporting scalable data warehouse architecture.
- Built automated data processing workflows and optimized SQL transformations, improving data quality, consistency, and reporting accuracy.
- Implemented analytics dashboards and KPI reporting using curated datasets, providing actionable insights into revenue trends, customer behavior, and business performance.
- Aligned the solution with modern cloud data warehouse practices using AWS services, demonstrating scalable architecture for enterprise analytics workloads.

EDUCATION

Master of Science in Information Systems and Technology

University of North Texas, Denton, TX

May 2025